

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I Prudhvi Raj.B(192324274) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prudhvi raj.b

Annexure A

Short Answer Test

1. A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.
2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.
3. A retail company wants to segment its customers based on purchasing behavior, but the data contains hidden patterns and incomplete information. Apply the EM algorithm to identify clusters and estimate underlying distributions. Analyze convergence behavior, initialization challenges, and the role of probability learning. Discuss how the results can be used for targeted marketing and evaluate limitations of the approach.
4. A research team is developing a concept learning system to classify medical images into normal and abnormal categories. Using hypothesis space search and the Candidate Elimination approach, design a model that learns from labelled examples. Analyze the impact of finite vs infinite hypothesis spaces, sample complexity, and mistake bound model on learning performance. Evaluate how well the system generalizes to unseen data.
5. A financial institution needs to make investment decisions under uncertain market conditions. Using Bayesian inference, Gibbs algorithm, and probability learning, design a system to predict outcomes and update beliefs dynamically. Analyze how prior knowledge influences predictions, evaluate risk using posterior probabilities, and discuss how the model adapts as new data becomes available. Consider computational challenges and optimization strategies.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Case Study 1

Probabilistic disease-diagnosis system using Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks.

1. System Design Overview

The hospital's diagnostic system takes patient symptoms and medical history as input evidence and outputs a probability distribution over possible diseases, rather than a single rigid answer. This is achieved by layering three probabilistic tools of increasing sophistication: Bayes' Theorem for basic evidence updating, a Naïve Bayes classifier for fast multi-symptom diagnosis, and a Bayesian Belief Network (BBN) to correctly model dependencies between symptoms.

2. Applying Bayes' Theorem

$$P(\text{Disease} \mid \text{Symptoms}) = [P(\text{Symptoms} \mid \text{Disease}) \times P(\text{Disease})] / P(\text{Symptoms})$$

Here, $P(\text{Disease})$ is the **prior** probability of the disease in the population, $P(\text{Symptoms} \mid \text{Disease})$ is the **likelihood** of observing the given symptoms if the patient has that disease, and $P(\text{Disease} \mid \text{Symptoms})$ is the **posterior** probability the system reports to the clinician. For example, if 2% of patients have a disease D (prior = 0.02), and 90% of patients with D show a fever (likelihood = 0.9), while 20% of all patients show a fever ($P(\text{Symptoms}) = 0.2$), then $P(D \mid \text{fever}) = (0.9 \times 0.02) / 0.2 = 0.09$, i.e., a 9% chance of disease D given fever alone — illustrating how Bayes' theorem revises belief in light of evidence.

3. Naïve Bayes Classifier Design

For fast screening across multiple symptoms, a Naïve Bayes classifier assumes symptoms are **conditionally independent given the disease**: $P(\text{Disease} \mid S_1, S_2, \dots, S_n) \propto P(\text{Disease}) \times \prod_{i=1}^n P(S_i \mid \text{Disease})$

The system estimates each $P(S \mid \text{Disease})$ and $P(\text{Disease})$ from historical patient records (probability learning, discussed below), then, for a new patient, multiplies these probabilities across all reported symptoms for every candidate disease and picks the disease with the highest posterior. This is computationally cheap and scales well even with many symptoms.

4. Bayesian Belief Network for Conditional Dependencies

The independence assumption of Naïve Bayes is unrealistic in medicine — symptoms such as *fever* and *chills*, or *cough* and *breathlessness*, are often correlated given the same underlying disease. A Bayesian Belief Network represents these relationships explicitly as a directed acyclic graph, where each node stores a Conditional Probability Table (CPT) that quantifies its dependence only on its direct parents, rather than assuming full independence.

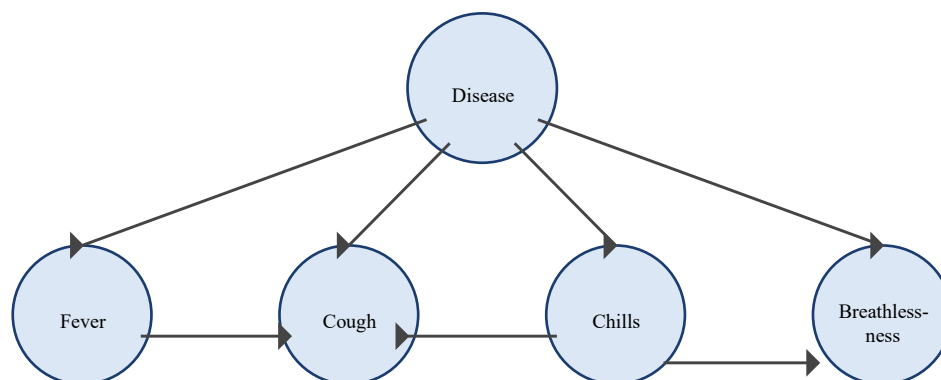


Fig. 1: Simplified Bayesian Belief Network – Disease node with symptom nodes, including direct edges between correlated symptoms.

Inference in the BBN combines the prior on Disease with the CPTs along the graph structure, so correlated symptoms reinforce (or dampen) each other's evidence appropriately instead of being treated as independent votes — giving more medically realistic posterior probabilities than plain Naïve Bayes.

5. Handling Uncertainty

All three tools output a **probability distribution** rather than a hard yes/no label, allowing the system to express confidence levels (e.g., “72% likely disease A, 20% disease B”), handle missing or unreported symptoms by marginalizing over them, and combine evidence incrementally as new test results arrive — the posterior from one stage becomes the prior for the next, refining the diagnosis as more information is gathered.

6. Role of Probability Learning

The prior probabilities $P(\text{Disease})$ and conditional probabilities $P(\text{Symptom} \mid \text{Disease})$ or the CPTs of the BBN are not hand-coded; they are **learned from historical patient records** using maximum likelihood estimation (relative frequency counts), typically with Laplace/additive smoothing to avoid zero probabilities for symptom-disease combinations that are rare but not impossible. As more patient data is collected, these learned probabilities are periodically re-estimated, allowing the system to improve over time.

7. Evaluation Against Real-World Constraints

- **Computational cost:** Exact inference in a general BBN is NP-hard as the network grows; large hospital systems with many symptoms/diseases may need approximate inference (e.g., sampling methods) to remain responsive.
- **Data sparsity for rare diseases:** Rare conditions have few historical cases, making learned probabilities unreliable; smoothing and expert-informed priors help mitigate this.
- **Independence assumption trade-off:** Naïve Bayes is fast but can misjudge cases with strongly correlated symptoms; the BBN is more accurate but costlier to build (requires domain expertise to define structure) and to run.
- **Interpretability and trust:** Clinicians need to understand why a probability was assigned; BBNs are more interpretable than opaque models, which matters for clinical adoption and accountability.
- **Data privacy and regulation:** Patient data used for probability learning must be handled under healthcare privacy regulations, constraining how data can be collected, stored, and reused.

Overall assessment: A layered approach — Naïve Bayes for quick triage and a BBN for cases needing dependency-aware reasoning — offers a practical balance between speed, accuracy, and interpretability, provided sufficient quality data is available to learn reliable probabilities and rare-disease handling is specifically addressed.

Case Study 2

Spam detection system using Naïve Bayes and the Bayes Optimal Classifier; MLE vs. MDL for feature selection.

1. System Design

Each email is represented as a bag-of-words feature vector (presence/frequency of vocabulary terms).

The classifier estimates, for each candidate class $c \in \{\text{spam}, \text{not-spam}\}$:

$$P(c \mid w_1, \dots, w_n) \propto P(c) \times \prod_i P(w_i \mid c)$$

$P(c)$ and $P(w \mid c)$ are learned from a labelled training set of emails using relative frequency counts i (with Laplace smoothing so unseen words don't zero out the whole product). A new email is classified as spam if $P(\text{spam} \mid \text{words}) > P(\text{not-spam} \mid \text{words})$.

2. Bayes Optimal Classifier

Rather than committing to a single “best” hypothesis (e.g., one particular trained Naïve Bayes model), the **Bayes Optimal Classifier** combines the predictions of every hypothesis in the hypothesis space, weighted by its posterior probability given the training data:

$$P(c \mid x) = \sum_h P(c \mid x, h) \times P(h \mid \text{Data})$$

This is provably the best possible classifier on average (minimizes expected classification error) given the hypothesis space and prior, but it is computationally expensive since it requires summing over all hypotheses. In practice, spam filters approximate it using a single high-posterior model (like Naïve Bayes) or a small ensemble, trading optimality for tractability.

3. MLE vs. MDL in Feature (Vocabulary) Selection

Aspect	Maximum Likelihood Estimation (MLE)	Minimum Description Length (MDL)
Objective	Choose parameters/features that maximize the likelihood of the observed training data	Choose the hypothesis that minimizes (length of hypothesis) + (length of data encoded using hypothesis)
Complexity penalty	None — favours whichever model fits the training data best, however complex	Explicitly penalizes complex hypotheses/ large feature sets (Occam's razor)
Risk	Tends to overfit by including many rare/ noisy words as informative features	Naturally prunes rare, low-value words that don't meaningfully compress/explain the data
Best suited when	Training data is abundant relative to feature/vocabulary size	Feature space is large relative to available data, or interpretability/generalization matters

For spam detection with a very large vocabulary, pure MLE-based feature selection risks treating rare words (which may appear in only one or two training emails) as strong spam/ham indicators purely by chance. MDL-based selection favours a more compact feature set that still explains the training data well, reducing overfitting and generally improving performance on new, unseen emails.

4. System Performance Analysis

Spam filters are typically evaluated using precision (fraction of emails flagged as spam that are truly spam) and recall (fraction of actual spam correctly caught), rather than raw accuracy alone, because the cost of a **false positive** (a genuine email marked as spam) is usually much higher than a false negative (a spam email that slips through). The system should therefore be tuned to favour high precision, even at some cost to recall.

5. Trade-offs in Model Complexity

- A simple Naïve Bayes model with a reduced, MDL-selected feature set trains fast, requires less data, and is easy to update, but may miss subtle word-combination cues that indicate spam.
- A richer model (larger vocabulary, n-grams, or a full Bayes Optimal combination of hypotheses) can capture more nuance but needs substantially more training data to estimate reliably and is far more computationally expensive to train and run at scale (millions of emails per day).

6. Effect of Sample Size and Hypothesis Space on Accuracy

With a small training sample, probability estimates $P(w | c)$ are noisy and a large hypothesis space $\mathcal{H}$ (many possible feature combinations) makes it easy to find a hypothesis that fits the training data well by chance but generalizes poorly — consistent with the general principle that the number of training examples needed to learn reliably grows with the size (or complexity) of the hypothesis space. As sample size grows, MLE estimates converge toward the true underlying probabilities (law of large numbers), narrowing the gap between MLE and MDL-based performance; with limited data, MDL's complexity penalty gives it a clear generalization advantage over unconstrained MLE.

Case Study 3

Customer segmentation with hidden patterns and incomplete data using the EM algorithm.

1. Modelling Approach

Each customer is represented by a feature vector of purchasing behaviour (e.g., purchase frequency, average spend, category preferences). Since the “true” customer segment is not observed, it is modelled as a **hidden (latent) variable**, and the overall population is modelled as a mixture of K probability distributions (e.g., a Gaussian Mixture Model), one per hidden segment, each with its own mean behaviour profile and mixing weight.

2. Applying the EM Algorithm

Step	Description
Initialization	Choose the number of segments K and initialize each segment's parameters (mean, covariance, mixing weight), e.g., via random draws or k-means++.
E-step (Expectation)	For every customer, compute the posterior probability (“responsibility”) of belonging to each segment, given the current parameter estimates and using Bayes' theorem: $r_{ik} \propto \pi_k \times P(x_i \text{segment } k)$.
M-step (Maximization)	Re-estimate each segment's mean, covariance, and mixing weight as a weighted average over all customers, weighted by the responsibilities computed in the E-step (a weighted maximum-likelihood update).
Iterate	Repeat E-step and M-step until the log-likelihood of the data under the model changes by less than a small threshold (convergence).

3. Convergence Behaviour

EM is guaranteed to **never decrease** the data log-likelihood at each iteration, so the algorithm reliably converges — but only to a **local** maximum of the likelihood, not necessarily the global optimum. Convergence is typically monitored by tracking the log-likelihood curve across iterations; it should rise quickly at first and then flatten as the model settles.

4. Initialization Challenges

- Poor random initialization can cause EM to converge to a low-quality local optimum, or to a degenerate solution where a cluster collapses onto a single customer (empty/near-empty cluster).
- The correct number of segments K is not known in advance; it is often chosen using model-selection criteria such as BIC or AIC, or by inspecting how log-likelihood improvement diminishes as K increases (an “elbow” method).
- Mitigation: run EM multiple times from different random initializations (or k-means++ style initialization) and keep the run with the highest final log-likelihood.

5. Role of Probability Learning

Rather than assigning each customer to exactly one segment (hard clustering), EM performs **soft, probabilistic clustering** — learning a full distribution over segment membership for every customer. This naturally handles **incomplete information**: if some behavioural attributes are missing for a customer, the E-step can marginalize over the missing values while still computing meaningful responsibilities, instead of discarding that customer's record.

6. Use for Targeted Marketing

Once converged, each mixture component represents an interpretable customer segment (e.g., “frequent low-spend buyers,” “occasional high-value buyers”). Marketing campaigns, offers, and product recommendations can then be tailored per segment, and the responsibility scores allow campaigns to be targeted proportionally even for customers who sit between segments (e.g., 70% likely segment A, 30% segment B).

7. Limitations

- Sensitive to initialization and may converge to a local, not global, optimum.
- Assumes a particular distributional shape (e.g., Gaussian) for each segment, which may not match real purchasing-behaviour distributions (often skewed or multi-modal).
- Choosing the number of segments K is subjective and requires additional model-selection steps.
- Computationally expensive for very large customer bases or high-dimensional behavioural features, and can converge slowly if segments overlap significantly.

Case Study 4

Concept learning for medical image classification using hypothesis space search and the Candidate Elimination algorithm.

1. Problem Setup

Each medical image is described by a set of discrete attributes (e.g., Shape $\in$ {Regular, Irregular}, Texture $\in$ {Smooth, Rough}, Density $\in$ {Low, High}, Border $\in$ {Well-defined, Diffuse}), and the target concept is “Abnormal” (positive) vs. “Normal” (negative). The learning system searches a hypothesis space H of conjunctive rules over these attributes (plus “?” for “any value” and $\emptyset$ for “no value”) to find a hypothesis consistent with all labelled training images.

2. Candidate Elimination Algorithm

The algorithm maintains two boundary sets that together define the **version space** — the set of all hypotheses consistent with the training examples seen so far:

- **Specific boundary (S)**: the most specific hypothesis(es) consistent with the data, initialized to the maximally specific hypothesis ($\emptyset, \emptyset, \dots$).
- **General boundary (G)**: the most general hypothesis(es) consistent with the data, initialized to the maximally general hypothesis ($?, ?, \dots$).

For each new training image: a **positive** example (Abnormal) generalizes S just enough to cover it, and removes from G any hypothesis inconsistent with it; a **negative** example (Normal) specializes G just enough to exclude it, and removes from S any hypothesis inconsistent with it. After processing all training images, every hypothesis strictly between S and G (in the generality ordering) is consistent with all observed data and forms the final version space.

3. Finite vs. Infinite Hypothesis Spaces

With discrete-valued attributes as above, the hypothesis space is **finite**, so Candidate Elimination can exhaustively and efficiently track the boundary of consistent hypotheses. If image features were continuous (e.g., raw pixel intensities or texture measures on a real-valued scale), the hypothesis space becomes **infinite**, and exhaustive boundary-based search is no longer feasible — this is why real medical-imaging systems typically move to statistical/gradient-based learners (e.g., CNNs) once features become continuous or very high-dimensional, while concept learning remains valuable for smaller, interpretable, discretized feature sets.

4. Sample Complexity

Within the PAC (Probably Approximately Correct) learning framework, the number of training examples needed to guarantee, with confidence $(1-\delta)$, that the learned hypothesis has error at most ϵ , grows with the size (or VC dimension) of the hypothesis space — roughly proportional to $(1/\epsilon)(\ln|H| + \ln(1/\delta))$ for finite H . A larger, more expressive hypothesis space (e.g., allowing more attribute combinations) can represent more complex abnormality patterns but requires proportionally more labelled medical images to learn reliably.

5. Mistake Bound Model

If the system learns **online**, updating its hypothesis as each new labelled image arrives, the mistake-bound model measures the worst-case number of misclassifications made before the hypothesis converges to the true target concept. For a finite hypothesis space, algorithms such as the Halving Algorithm guarantee at most $\log_2|H|$ mistakes — giving a theoretical upper bound on how many misdiagnoses the system might make while still learning, which is a useful safety-relevant metric in a medical setting.

6. Generalization to Unseen Data

Candidate Elimination is guaranteed to converge to the exact target concept only if (a) the true concept lies within the chosen hypothesis space, and (b) the training data is noise-free. In practice, medical images contain noisy, ambiguous, or borderline cases, which can cause the version space to collapse to the empty set (no consistent hypothesis) or leave an overly broad version space that generalizes poorly. Consequently, while the approach gives clear, interpretable, and provably-consistent rules on clean, well-structured feature data, its generalization to the full complexity and noise of real clinical images is limited compared to statistical learners that tolerate noise and learn probabilistic decision boundaries.

Case Study 5

Investment decision-making under uncertainty using Bayesian inference, the Gibbs algorithm, and probability learning.

1. System Design

The system treats each possible market outcome (e.g., asset price movement, return range) as a random variable, and maintains a **belief distribution** over these outcomes that is updated as new market data (earnings reports, price movements, economic indicators) arrives, using Bayes' theorem:

$$P(\text{Outcome} \mid \text{Data}) \propto P(\text{Data} \mid \text{Outcome}) \times P(\text{Outcome})$$

2. Role of Prior Knowledge

The **prior** $P(\text{Outcome})$ encodes existing beliefs before seeing new data — for example, historical average returns, analyst expectations, or macroeconomic trends. When data is scarce (e.g., a newly listed asset), the prior dominates the posterior; as more market data accumulates, the likelihood term increasingly dominates and the posterior converges toward what the data itself indicates, gradually reducing the influence of the initial prior (a property of Bayesian updating as sample size grows).

3. Gibbs Sampling for Complex Posteriors

Financial models typically involve many interacting variables (multiple assets, macro indicators, correlated risk factors), making the joint posterior distribution analytically intractable. The **Gibbs sampling** algorithm approximates this posterior by iteratively sampling each variable from its conditional distribution given the current values of all other variables, cycling through all variables repeatedly. After a “burn-in” period, the resulting samples approximate draws from the true joint posterior, from which any quantity of interest (expected return, probability of loss, etc.) can be estimated by averaging over the samples.

4. Evaluating Risk Using Posterior Probabilities

Once the posterior distribution (or its Gibbs-sampled approximation) is available, risk metrics can be computed directly — for instance, the probability that returns fall below a critical threshold, $P(\text{Return} < -x \mid \text{Data})$, gives a Bayesian analogue of Value-at-Risk, and the full posterior can be used to compute expected utility or expected loss under different investment choices, allowing decisions to explicitly account for uncertainty rather than relying on a single point forecast.

5. Dynamic Belief Updating

The system is designed for **sequential Bayesian updating**: the posterior computed after observing data up to time t is used as the prior for time $t+1$, so beliefs about market conditions are continuously refined as new data streams in, without needing to recompute everything from scratch — well suited to live, evolving market conditions.

6. Computational Challenges and Optimization Strategies

Challenge	Optimization Strategy
High-dimensional, intractable joint posterior over many market variables	Use Gibbs sampling / MCMC to approximate the posterior instead of exact computation
Slow mixing / convergence of sampling chains	Run multiple parallel chains; use good starting values; monitor convergence diagnostics
Repeated sampling too slow for real-time trading decisions	Use conjugate priors (e.g., Normal-Normal, Beta-Binomial) for closed-form posterior updates where possible, avoiding sampling entirely
Full re-sampling needed as new data continuously arrives	Use sequential/online methods such as particle filters, or variational inference, for faster approximate updates than full Gibbs re-runs

Overall: Combining Bayesian inference for principled belief updating, Gibbs sampling for handling otherwise-intractable multi-variable posteriors, and conjugate/online approximations where speed is critical, allows the institution to make investment decisions that explicitly quantify and adapt to uncertainty, while keeping computation tractable enough for practical, near-real-time use.